



SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

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Wait-Notify

WAIT-NOTIFY

Write a Java program to demonstrate how to use the "wait" and "notify" methods to implement a producer-consumer scenario.

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {  
2  
3     // Shared Buffer  
4     static class Buffer {  
5         private final int[] items;  
6         private int count = 0;  
7         private int nextIndex = 0;  
8  
9         Buffer(int capacity) {  
10             items = new int[capacity];  
11         }  
12  
13         public synchronized void put(int value) throws InterruptedException {  
14             while (count == items.length) {  
15                 System.out.println("Buffer full. Producer waits...");  
16                 wait();  
17             }  
18             items[nextIndex] = value;  
19             nextIndex = (nextIndex + 1) % items.length;  
20             count++;  
21
```

OUTPUT

```
Produced: 1 | Buffer count: 1  
Consumed: 1 | Buffer count: 0  
Produced: 2 | Buffer count: 1  
Consumed: 2 | Buffer count: 0  
Produced: 3 | Buffer count: 1  
Consumed: 3 | Buffer count: 0  
Produced: 4 | Buffer count: 1  
Produced: 5 | Buffer count: 2  
Consumed: 4 | Buffer count: 1  
Produced: 6 | Buffer count: 2
```

**INPUT**

Your INPUT goes here!

